Hazem Hassan

AI Engineer – LLM & Agentic Systems – Production ML

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PROFESSIONAL SUMMARY

Senior AI Engineer with 9+ years building and shipping production AI systems end-to-end — not notebooks. Recent focus on LLM-powered applications: agentic workflows with LangGraph, RAG systems with LlamaIndex, and a full LLM evaluation/observability stack (Ragas, LangSmith). Strong software-engineering foundation — production Python, clean FastAPI/REST API design, robust error handling, Docker/Kubernetes, CI/CD — with a habit of thinking in latency, cost-per-query, and output-quality metrics. Comfortable owning the full lifecycle: architecture, code, deployment, evaluation, and monitoring.

ROLES AND SPECIALTIES

Roles: Senior ML Engineer, AI Technical Lead, LLM/Agentic Systems Engineer.

Specialties: Agentic Workflows, RAG, LLM Evaluation, Automated Labeling, Human-in-the-Loop Workflows, Active Learning, Weak Supervision, Dataset Quality Assurance, ML APIs, MLOps, Cloud/Edge Deployment.

CORE SKILLS

LLM & Agentic Systems: LangGraph (agent workflows / tool-calling), LlamaIndex (RAG, embedding fine-tuning, retrieval orchestration), Multi-RAG, Embeddings, Reranking, Chunking Strategies, FAISS, Conversation Orchestration, NeMo Guardrails, Human-in-the-Loop / Escalation Routing, Prompt Engineering

LLM Evaluation & Observability: Ragas (component-level retriever/generator eval), LangSmith (tracing + continuous evaluation), MT-Bench (LLM-as-judge), AlpacaEval, LM-Eval Harness, Elo Rating, Perplexity, Concurrency/Latency Benchmarking

Software Engineering: Production Python (expert), Clean REST/API Design (FastAPI), Error Handling, Modular Service Design, Multi-threaded Real-time Systems, C/C++ (intermediate), Git, Linux

Deployment & MLOps: Docker, Kubernetes, CI/CD, Model Serving, Inference Optimization (ONNX, TensorRT), Monitoring & Failure Analysis

Cloud & Data: GCP (Vertex AI), AWS (SageMaker), SQL / PostgreSQL, NoSQL basics, Pandas, NumPy, Large-Scale Data Pipelines

ML/DL: PyTorch, TensorFlow, Scikit-learn, Model Training & Fine-Tuning, Hyperparameter Tuning, Custom Loss Design, Evaluation Metrics, Error Analysis.

Leadership: Technical Leadership, System Design, Cross-functional Collaboration, Mentoring, Code Reviews.

WORK EXPERIENCE

Senior ML Engineer

Beyond AI — March 2024—Present, Remote, EMEA

- Built agentic LLM workflows with LangGraph — modeling agents as directed graphs with message passing, tool-calling, and execution orchestration across nodes — plus custom orchestration and RAG services on Python + FastAPI.
- Developed RAG systems with LlamaIndex: embedding fine-tuning, retrieval orchestration, synthetic QA generation, and multi-modal pipelines over text, charts, and research papers, backed by a FAISS vector store.
- Engineered a long-document RAG path (recursive + sequential summarization) with NeMo Guardrails intent routing between retrieval and summarization, tuning chunk size to hold time-to-first-token under 5 seconds.
- Built an LLM evaluation framework integrating Ragas (component-level retriever/generator scoring — context relevancy/recall, faithfulness, answer relevancy) and LangSmith for tracing and continuous evaluation; added MT-Bench (LLM-as-judge), AlpacaEval, LM-Eval, Elo, perplexity, and a concurrency/latency benchmark measuring per-token timing under load.
- Prototyped an agentic AI call-center assistant: LLM tool-calling + Multi-RAG for intent handling and answer generation with CRM/ticketing actions, guardrails, human-in-the-loop escalation, and an output-quality evaluation loop.

Technical Lead

IDefy (Digital Identity/eKYC) — October 2022—February 2024, Giza, Egypt

- Led digital identity onboarding products (SaaS/SDK/web/mobile), delivering 3 production deployments end-to-end.
- Built validation and quality-control workflows for OCR, face matching, liveness, and identity-verification datasets, improving verification success rate by 20% through data quality tuning and systematic failure analysis.
- Developed configurable OCR/document-processing pipelines that adapted to new structured documents using limited samples, annotation rules, and validation checks, reaching 96% recognition accuracy.
- Analyzed production failure cases across OCR, face verification, and liveness detection to identify noisy data, poor image quality, edge cases, and model retraining opportunities.

Senior ML Engineer (Computer Vision)

RDI — January 2021—October 2022, Giza, Egypt

- Led the delivery of the Sotoor Arabic OCR System, achieving 12% WER on Arabic manuscripts.
- Built document OCR pipeline modules (layout analysis, segmentation, font detection, recognition, denoising, correction), reducing OCR error by ~7% on noisy scans.
- Adapted the K2 Toolkit for OCR training/decoding, improving training efficiency by ~25% and gaining ~4% WER, boosting sequence recognition stability.

ML Researcher

RDI — January 2017—January 2021, Giza, Egypt

- Collaborated on a team building NLP systems for the Arabic language across speech and text.
- Developed and deployed production systems, including Speech Recognition, Optical Character Recognition, Text Diacritization, Voice Conversion, and Quran Recitation Verification.

PROJECTS

Agentic Agent Call Center

- Built an agentic AI call-center assistant using LLM tool-calling and Multi-RAG retrieval to automate intent handling, answer generation, and CRM/ticketing actions.
- Designed a reliable conversation orchestration layer with guardrails, escalation-to-human routing, and quality evaluation to ensure safe and consistent customer support automation.

Fashimi APP — Lead Engineer

- Built an AI try-on platform using LLMs for recommendation + prompt generation and diffusion models for avatar generation and photorealistic try-on rendering.
- Developed a recommendation engine leveraging mood/occasion/climate/user preferences to produce personalized outfit prompts.

Valorant Overlay — Developer

- Designed a real-time multi-threaded system extracting gameplay data from Valorant for esports competition overlays.
- Implemented object detection, pattern recognition, and color analysis, achieving a response latency of 50-72 msec.

EDUCATION

Bachelor of Science in Engineering

Cairo University – Electrical Electronics and Communication Engineering · 2016

PUBLICATIONS

Arabic Multi-Genre Broadcast Speech Recognition — Conference Paper, December 2019 — IEEE ASRU 2019

Prognostic AI for Post-Operative Infections — Conference Paper, November 2023 — FCI Annual Conference

QuranScript Verification System — Conference Paper, November 2016 — Sixteenth ESOLE Conference of Language Engineering

HOBBIES

Swimming | Gaming | Padel